

# Supplementary Material for: Robust Branch-Safe Participation for Sequential Mobile Crowdsensing

Lili He<sup>a</sup>, Sheng Jiang<sup>a</sup>, Linghui Lyu<sup>a</sup>, Lei Zhang<sup>a,\*</sup>

<sup>a</sup>*College of Information and Electronic Technology, Jiamusi University, Jiamusi 154007, China*

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## Abstract

This supplement provides extended implementation details, complete diagnostic plots, comparator adaptations, model-mismatch stress tests, task-size analysis, and reproducibility information supporting the main manuscript. It does not introduce additional primary claims. The post-hoc status of the robustness extensions and all negative or boundary results are preserved.

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## 1. Extended observation-channel example

For a predicted belief  $b = (0.2, 0.2, 0.2, 0.4)$ , task region  $S = \{1, 2, 3\}$ , availability  $\alpha = 0.9$ , and local cap  $c = 0.4$ , the truthful report posterior has peak  $1/3$ , so a success-only check would permit full participation. If  $q = 1$ , however, silence leaves the fourth state with posterior probability  $0.4/(1 - 0.9 \times 0.6) \approx 0.870$ , violating the cap. BSP therefore rejects this task under the two-branch constraint. The example isolates the role of probabilistic silence without using simulator ground truth.

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\*Corresponding author.

*Email address:* 8213662@163.com (Lei Zhang)

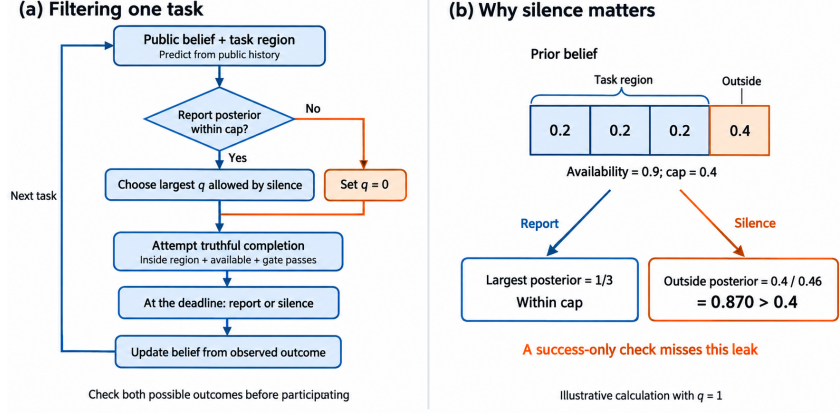


Figure 1: Original BSP workflow and analytical report/silence example.

## 2. Comparator adaptations and validation

### 2.1. Pointwise maximal leakage task channel

Let  $l_i = \Pr(Y = 1 \mid X = i)$  and  $m = \sum_i b_i l_i$ , with  $0 \leq l_i \leq \alpha \mathbf{1}\{i \in S\}$ . The task-channel PML specialization maximizes report probability subject to the published pointwise leakage inequalities  $l_i \leq e^\epsilon m$  and  $1 - l_i \leq e^\epsilon (1 - m)$  for every state with positive prior mass [1]. The implementation is checked against a general linear program over randomized instances. This is a channel specialization, not a reproduction of every component of the original application.

### 2.2. PRIVIC task adapter

PRIVIC learns a noisy location channel through Blahut–Arimoto and incremental Bayesian updates [2]. We check our BA/IBU/GIBU updates against the authors’ public implementation. The task adapter emits a truthful report only when the true state is eligible and the location-channel outcome satisfies the task predicate. Because this changes the output semantics, PRIVIC-T does not inherit the original geo-indistinguishability guarantee. Offline channel fitting is excluded from online gate timing.

### 2.3. Per-output KL comparator

The Bayesian-privacy comparator constrains each positive-probability posterior using  $D_{\text{KL}}(b^y||b) \leq \kappa$  [3]. The successful-report divergence is fixed by the task support; the silence bound is solved by bisection. It is a per-output diagnostic rather than a cumulative session guarantee.

### 3. Complete nominal and recent-method diagnostics

The main manuscript reports only the decision-relevant summary. Figures below retain the executed policy settings and exploratory envelope construction used during analysis. Envelope selection uses evaluation results and is therefore not presented as confirmatory tuning. At opportunity retention 0.5 in the static setting,

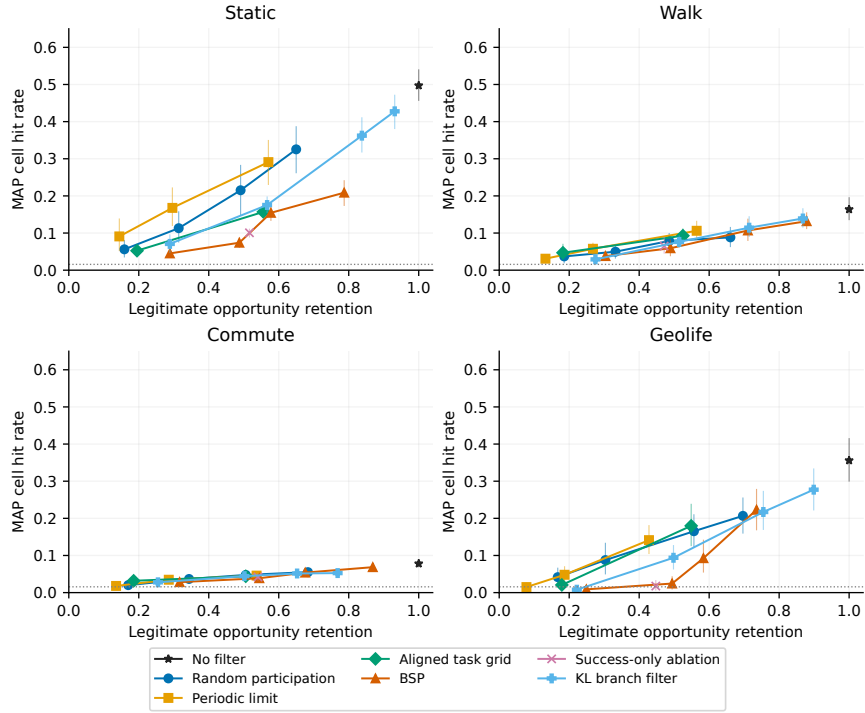


Figure 2: Original privacy-utility trade-off across the main policy grid.

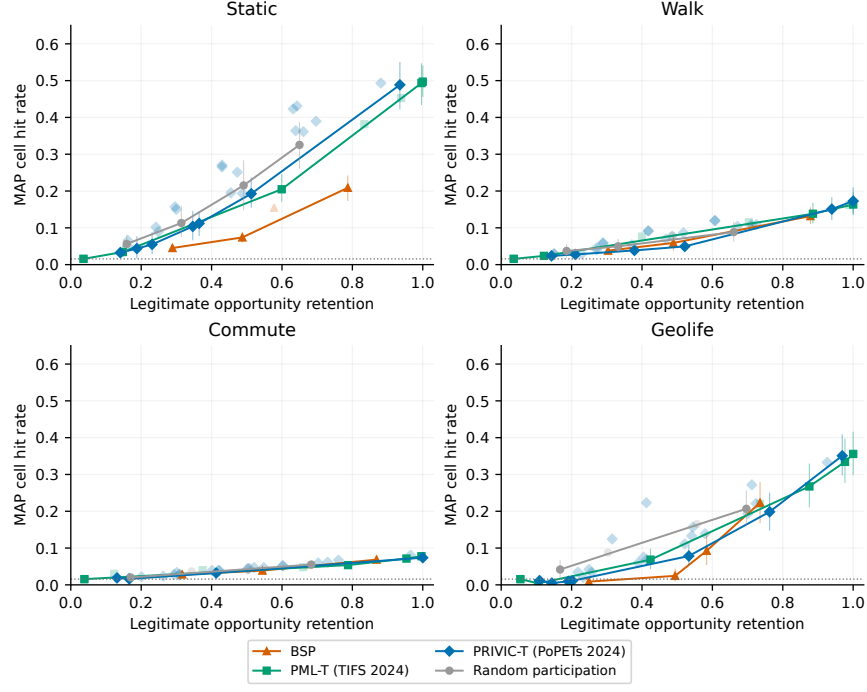


Figure 3: Exploratory comparison with PML-T and PRIVIC-T. Faint points are executed settings; curves are evaluation-selected lower envelopes.

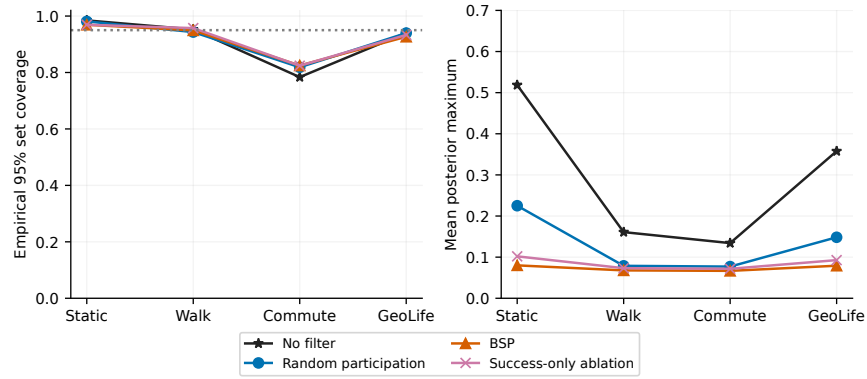


Figure 4: Calibration diagnostics for selected attacks and defenses.

exploratory interpolation gives BSP MAP hit rate about 7.96%, PML-T 16.74%, and PRIVIC-T 18.58%. Walk, commute, and GeoLife comparisons do not support a consistent superiority claim over both recent comparators. The development-fitted no-output modal predictor reaches 25.75% test accuracy, demonstrating that the uniform-prior replay attack is not a practical upper bound.

#### 4. Sensitivity and stronger attacker knowledge

The sensitivity analysis varies availability, mobility assumptions, deadline success, grid resolution, and idealized history reset. Ten seed clusters contain four users each. These analyses are exploratory and are not used to retune the main operating points. When the client assumes availability 0.3 while the informed attacker knows

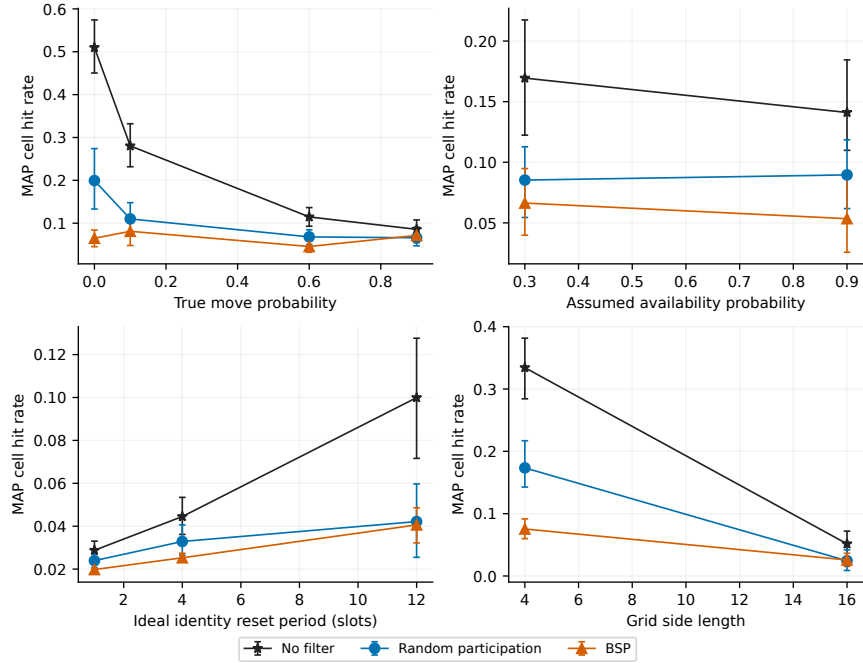


Figure 5: Selected sensitivity results with pointwise cluster-bootstrap intervals.

the generating  $\alpha = 0.648$ , the attacker's posterior peak exceeds 0.1 on 30.0% of

evaluated slots and reaches 0.246. When the client assumes move probability 0.8 while the attacker knows the generating value 0.3, the informed peak exceeds 0.1 on 41.4% of slots and reaches 0.263. In both cases the client-side belief can appear compliant, showing that nominal self-auditing is insufficient under model mismatch.

Table 1: Original informed-attacker stress table.

| Client mismatch      | Retention | MAP hit | Above cap | Largest peak |
|----------------------|-----------|---------|-----------|--------------|
| $\hat{\alpha} = 0.3$ | 0.551     | 0.098   | 30.0%     | 0.246        |
| $\hat{\alpha} = 0.9$ | 0.464     | 0.057   | 8.0%      | 0.164        |
| $\hat{\nu} = 0.05$   | 0.457     | 0.038   | 4.7%      | 0.153        |
| $\hat{\nu} = 0.8$    | 0.543     | 0.083   | 41.4%     | 0.263        |

## 5. Finite-model robustness stress

The hand-grid ambiguity set uses  $\nu \in \{0.05, 0.3, 0.8\}$  and  $\alpha \in \{0.3, 0.648, 0.9\}$ . Twenty new synthetic seeds, four users per seed, and 48 slots yield 80 trajectories per row at a fixed operating point. When the informed attacker’s exact generating model is contained in the set, R-BSP records no local-cap violation in the five reported conditions, as expected from the finite-set construction, while retaining about 39.7% of externally completable normal-task opportunities. A separate boundary study fixes the same set but evaluates correct attacker models that are not exact members. The outside-set value  $\alpha = 0.98$  yields a 6.93% local-cap violation rate with a cluster-bootstrap 95% interval of 5.70–8.31%. Interior unlisted values yield zero or rare violations in this finite sample. These measurements are why the main manuscript states a finite-model rather than universal robustness guarantee.

Table 2: Finite-model R-BSP stress at  $\rho = 0.1$ .

| Client model             | BSP viol. | R-BSP viol. | BSP retention | R-BSP retention |
|--------------------------|-----------|-------------|---------------|-----------------|
| Nominal                  | 0.0%      | 0.0%        | 50.9%         | 39.7%           |
| $\widehat{\alpha} = 0.3$ | 12.6%     | 0.0%        | 54.2%         | 39.7%           |
| $\widehat{\alpha} = 0.9$ | 2.6%      | 0.0%        | 47.3%         | 39.7%           |
| $\widehat{v} = 0.05$     | 2.8%      | 0.0%        | 46.9%         | 39.7%           |
| $\widehat{v} = 0.8$      | 14.6%     | 0.0%        | 55.2%         | 39.7%           |

Table 3: Boundary characterization for attacker models not explicitly enumerated in the hand-grid ambiguity set.

| Attacker model             | BSP viol. | R-BSP viol. | BSP ret. | R-BSP ret. |
|----------------------------|-----------|-------------|----------|------------|
| $\alpha = 0.50$ (interior) | 1.15%     | 0.00%       | 52.6%    | 36.3%      |
| $\alpha = 0.98$ (outside)  | 18.33%    | 6.93%       | 50.0%    | 40.4%      |
| $v = 0.50$ (interior)      | 0.65%     | 0.05%       | 54.4%    | 36.4%      |
| $v = 0.95$ (outside)       | 0.65%     | 0.03%       | 55.6%    | 35.4%      |

## 6. Task-size composition and disclosure floor

The representative strict BSP traces show a strong dependence on task area. Opportunity counts use simulator eligibility and the shared external availability draw; completion additionally requires admission. For a uniform prior over an

Table 4: Task-area breakdown for strict BSP at  $\rho = 0.1$ .

| Scenario | Area (cells) | Tasks | Opportunities | Completed | Retention |
|----------|--------------|-------|---------------|-----------|-----------|
| Static   | 1–8          | 2222  | 55            | 0         | 0.000     |
| Static   | 9–31         | 313   | 53            | 22        | 0.415     |
| Static   | 32–64        | 344   | 169           | 113       | 0.669     |
| Walk     | 1–8          | 2222  | 63            | 0         | 0.000     |
| Walk     | 9–31         | 313   | 53            | 13        | 0.245     |
| Walk     | 32–64        | 344   | 164           | 124       | 0.756     |
| Commute  | 1–8          | 2222  | 57            | 0         | 0.000     |
| Commute  | 9–31         | 313   | 45            | 22        | 0.489     |
| Commute  | 32–64        | 344   | 157           | 119       | 0.758     |
| Geolife  | 1–8          | 1829  | 49            | 0         | 0.000     |
| Geolife  | 9–31         | 274   | 59            | 18        | 0.305     |
| Geolife  | 32–64        | 281   | 149           | 109       | 0.732     |

$m$ -cell region, a truthful positive report has posterior peak  $1/m$ . Hence a strict  $\rho = 0.1$  report from fewer than ten cells cannot be made feasible by changing a scalar thinning probability. This structural floor motivates the asymmetric DF/RDF mechanisms rather than another threshold search.

## 7. Data-driven ambiguity construction

The replay ambiguity set is built without using the 67 test users. Twenty-three development users define a coarsened reflecting-walk likelihood  $\Pr(\text{change} \mid i) =$

$vd_i/4$  and a development-only smoothed population prior; only change versus no-change events enter this likelihood, not destination direction or jump distance. Two thousand user-level development bootstraps produce mobility candidates, and 2,000 validation-user bootstraps select the smallest set covering at least 95% of predictive-likelihood winners. The first five candidates account for about 93.65% of validation-bootstrap winners, so all six population-prior mobility candidates are retained, giving 100% achieved support in this finite candidate set. The development MLE is 0.0514 with a 95% bootstrap interval 0.0318–0.0734; the validation point estimate is 0.0573. Availability remains externally specified because GeoLife contains trajectories rather than task-delivery, willingness, or deadline logs. A validation-best single-model BSP control separates improved model fit from finite-set robustness. Under the stronger replay attacker, its local-cap violation rate is 2.9%, compared with zero observed local-cap violations for strict data-driven R-BSP, but the latter retains only 5.4% of opportunities.

## 8. Asymmetric disclosure-floor frontier

DF-BSP relaxes only the truthful report branch. Reports use  $c_{\text{DF}}^+ = \max\{\tau, \|b\|_\infty\}$  while silence retains  $c_{\text{DF}}^- = \max\{\rho, \|b\|_\infty\}$ . At  $\tau = \rho$  the rule reduces exactly to BSP. Randomized tests check report admissibility, strict silence feasibility, robust member-wise intersection, and scalar maximality. On new synthetic walks,  $\tau = 0.5$  raises 1–8-cell retention from zero to 56.4% and inverse-area-weighted retention from 12.3% to 43.1%, with no observed declared branch-cap violation. In the stronger GeoLife replay, non-robust DF-BSP has a 16.4% branch-cap violation rate at the same  $\tau$ , whereas data-driven RDF-BSP reduces it to 0.16% while retaining 14.3% of small-region opportunities and 20.8% inverse-area-weighted utility. The original BSP local cap is violated on 14.3% of slots because the relaxed report-side

threshold can permit greater posterior concentration on truthful reports.

## 9. Expanded commute view

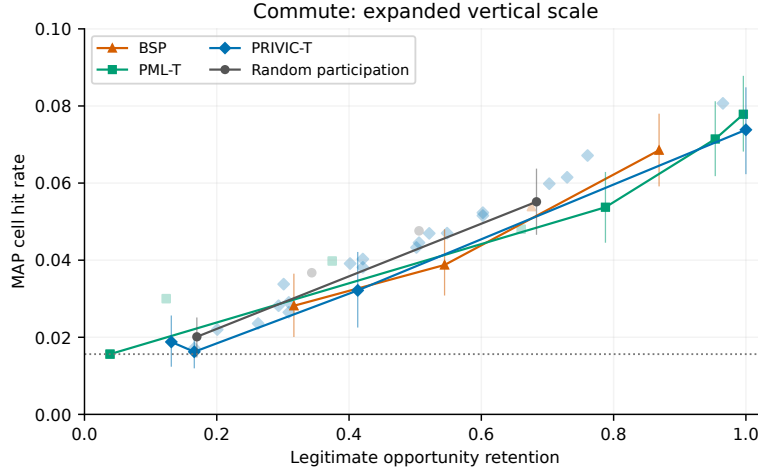


Figure 6: Expanded commute comparison using the same saved estimates and intervals as the recent-method analysis.

The expanded view changes only the vertical scale. It does not alter the conclusion that the commute comparison is inconclusive.

## 10. Reproducibility record

The repository stores deterministic preprocessing, split manifests, configuration files, seeds, event-level compressed logs, trajectory-level summaries, bootstrap outputs, and provenance records. The current automated suite contains 28 tests covering nominal, robust, and disclosure-floor posterior branches; gate reductions and maximality; informed-attacker integration; transition conservation; GeoLife split/missing-minute handling; numerical replay; PML linear-program

checks; PRIVIC author-code checks; and policy-envelope optimization. The GeoLife test split is disjoint from the 23 development and 17 validation users. The data-driven extension is nevertheless labelled post-hoc because the test replay had already been inspected earlier in the project. No mobile-device energy measurement, production MCS task log, or independent external reproduction is claimed.

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